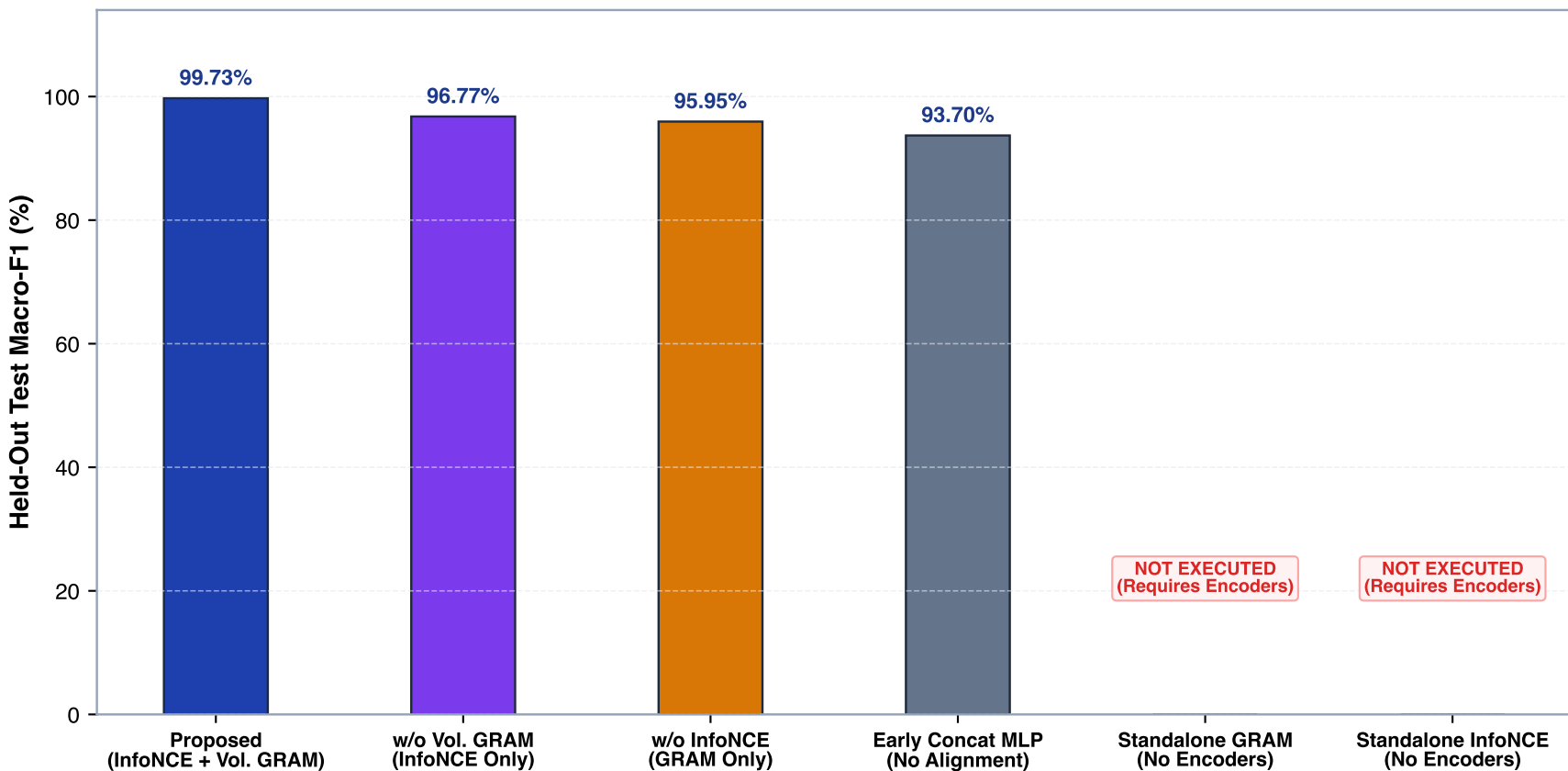


# Contrastive Hyperspherical Alignment Ablation (InfoNCE vs. Volumetric GRAM)



Note: Loss functions require modality encoder backbones to evaluate representation learning; standalone loss-only variants are marked NOT\_EXECUTED under zero-fabrication policy.